

10. Digital electronics

Yamulapalli Nandu Navadeep

Roll Number: IMS24267

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1 Aim of the Experiment

1. To verify the truth table of a BCD to Decimal Decoder using IC 7442.
2. To design and set up a 4-bit R-2R ladder Digital to Analog Converter (DAC) and measure the analog output voltage for different binary inputs.

2 Theory

2.1 BCD to Decimal Decoder

A BCD (Binary-Coded Decimal) to decimal decoder is a combinational logic circuit that converts a 4-bit BCD input into one of ten possible decimal digits. The IC 7442 is a commonly used BCD to decimal decoder. It accepts four active-high BCD inputs and provides ten active-low mutually exclusive outputs. When a valid BCD code (0000 to 1001) is applied to the inputs, the corresponding output line goes low (logic 0), while all other output lines remain high (logic 1). For invalid BCD inputs (1010 to 1111), all outputs remain high.

2.2 Digital to Analog Converter (DAC)

A Digital to Analog Converter (DAC) converts a digital input signal into a corresponding analog output voltage. The R-2R ladder network is a popular DAC architecture that uses resistors of only two values: R and $2R$. This makes it easier to implement accurately since only the ratio of the resistors matters. The digital inputs are applied to the $2R$ resistors in the network. The analog output is proportional to the value of the binary input. The magnitude of the output analog voltage for a 4-bit R-2R DAC is given by the general expression:

$$V_a = V_H(2^{-1}a_1 + 2^{-2}a_2 + 2^{-3}a_3 + 2^{-4}a_4)$$

Where V_H is the voltage at the logic 1 state, and a_1, a_2, a_3, a_4 represent the binary states (0 or 1). The output is maximum when all the inputs are at logic high.

3 Procedure

3.1 BCD to Decimal Decoder

1. Connect the IC 7442 on the trainer kit.
2. Apply +5 V to V_{CC} (Pin 16) and connect ground to GND (Pin 8).
3. Apply the 4-bit binary input to pins 15, 14, 13, and 12 corresponding to A, B, C, and D respectively.
4. Connect the output pins (1 to 7, 9 to 11) to LEDs to observe the logic states.
5. Feed binary inputs from 0000 to 1111 and verify the active-low output for each valid BCD input. Observe that all outputs are high for invalid BCD inputs.

3.2 4-bit R-2R Ladder DAC

1. Set up the R-2R ladder network using 10 k Ω and 22 k Ω resistors with an operational amplifier (IC 741).
2. Provide the necessary power supply (+15 V and -15 V) to the IC 741.
3. Manually feed the 4-bit binary inputs from 0000 through 1111.
4. Measure the analog output voltage at the op-amp output for each binary input using a multimeter.
5. Record the measured values in a tabular column.

4 Circuit Diagrams

4.1 BCD to Decimal Decoder (IC 7442)

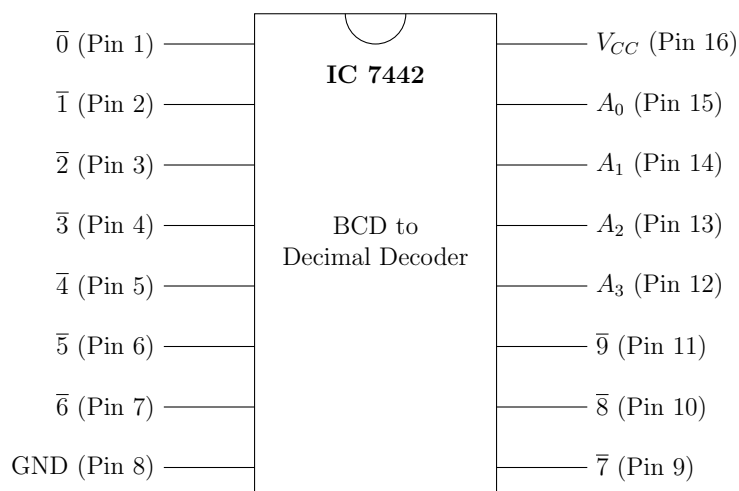


Figure 1: Pin diagram of BCD to Decimal Decoder (IC 7442)

4.2 4-bit R-2R Ladder DAC

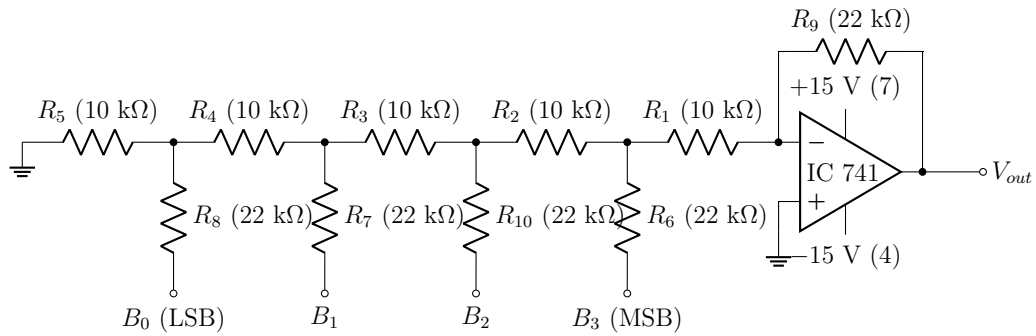


Figure 2: Circuit diagram of 4-bit R-2R Ladder DAC

5 Tabular Column

5.1 BCD to Decimal Decoder (IC 7442)

[illegible]

Table 1: Truth Table for BCD to Decimal Decoder (IC 7442)

5.2 4-bit R-2R Ladder DAC

Binary Input				Decimal Input	Measured Output Voltage (V)
B_3	B_2	B_1	B_0		
0	0	0	0	0	0
0	0	0	1	1	0.245
0	0	1	0	2	0.580
0	0	1	1	3	0.834
0	1	0	0	4	1.225
0	1	0	1	5	1.466
0	1	1	0	6	1.815
0	1	1	1	7	2.056
1	0	0	0	8	2.386
1	0	0	1	9	2.627
1	0	1	0	10	2.988
1	0	1	1	11	3.216
1	1	0	0	12	3.607
1	1	0	1	13	3.848
1	1	1	0	14	3.857
1	1	1	1	15	4.20

Table 2: Analog Voltage output for corresponding Binary Inputs

6 Calculations

The resolution of the DAC can be estimated by taking the difference in measured output voltage for consecutive binary inputs. For a 4-bit DAC, the theoretical step size is given by:

$$\text{Step Size} = \frac{V_{FS}}{2^n - 1}$$

Where V_{FS} is the full-scale voltage and $n = 4$. From our observations, the full scale voltage is approximately 4.20 V.

$$\text{Step Size} \approx \frac{4.20}{15} = 0.28 \text{ V}$$

This theoretical step size is consistent with the voltage increments observed between consecutive digital inputs, taking into account experimental uncertainties due to resistor tolerances.

7 Results

The truth table of the BCD to Decimal Decoder (IC 7442) was verified successfully, demonstrating active-low outputs for valid BCD inputs and high logic states for invalid inputs.

The 4-bit R-2R ladder Digital to Analog Converter was successfully set up. The analog output voltages for different binary inputs were measured and recorded. The output voltage increased proportionally with the binary input value.

8 Discussion

In this experiment, two fundamental digital electronic circuits were studied. First, the IC 7442 decoder was verified. It effectively converted a 4-bit BCD input into a decimal line by driving a specific output low. The observation that invalid BCD inputs (greater than 9) did not trigger any of the zero-to-nine output lines confirmed the designed behavior of the decoder.

The second part of the experiment involved building a 4-bit DAC using an R-2R ladder network and an operational amplifier. The R-2R ladder configuration proved advantageous as it required only two resistance values ($10\text{ k}\Omega$ and $22\text{ k}\Omega$), making the circuit straightforward to construct. The measured analog voltages showed a proportional relationship with the corresponding binary inputs. However, there were slight deviations from perfectly linear voltage steps. These deviations are primarily due to the tolerance values of the resistors used. For a more precise DAC, precision resistors with tight tolerances are required to minimize conversion errors.

9 References

1. Lab Manual for Advanced Physics Lab II, IISER TVM.

10 Lab Notes (Observations)

The following handwritten notes and observations were recorded during the laboratory session.

	A ₃	A ₂	A ₁	A ₀	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇	Y ₈	Y ₉
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	1	0	0	0	0	0	0	0
3	0	0	1	1	0	0	0	1	0	0	0	0	0	0
4	0	1	0	0	0	1	1	0	0	0	0	0	0	0
5	0	1	0	1	0	0	0	1	1	0	0	0	0	0
6	0	1	1	0	0	0	1	0	1	0	0	0	0	0
7	0	1	1	1	0	0	0	0	1	1	0	0	0	0
8	1	0	0	0	0	0	0	0	0	0	1	0	0	0
9	1	0	0	1	0	0	0	0	0	0	0	1	0	0
10	1	0	1	0	0	0	0	0	0	0	0	0	1	0
11	1	0	1	1	0	0	0	0	0	0	0	0	0	1
12	1	1	0	0	0	0	0	0	0	0	0	0	0	0
13	1	1	0	1	0	0	0	0	0	0	0	0	0	0
14	1	1	1	0	0	0	0	0	0	0	0	0	0	0
15	1	1	1	1	0	0	0	0	0	0	0	0	0	0

Invalid all 1's

1 1 0 1

0 0 1 1

...

0000

0001

0010

0011

0100

0101

0110

0111

1000

1001

1010

1011

1100

1101

1110

1111

17105126

1M524267

Figure 3: Lab Notes for BCD to Decimal Decoder

<u>Input (Binary)</u>	<u>Input (decimal)</u>	<u>V_{measured} (V)</u>
0 0 0 0	0	+ 0
0 0 0 1	1	+ 0.245
0 0 1 0	2	0.589
0 0 1 1	3	0.834
0 1 0 0	4	1.225
0 1 0 1	5	1.466
0 1 1 0	6	1.815 1.815
0 1 1 1	7	2.056
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1 1 0 0	12	3.607
1 1 0 1	13	3.848
1 1 1 0	14	3.857
1 1 1 1	15	4.20

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 RMS 24267

Figure 4: Lab Notes for R-2R Digital to Analog Converter